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Clinical elicitation of the penilo-cavernosus reflex in circumcised men

Simon Podnar

Institute of Clinical Neurophysiology, Division of Neurology, University Medical Centre, Ljubljana, Slovenia

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Study Type – Aetiology (case series)
Level of Evidence 4

What's known on the subject? and What does the study add?
It is known that foreskin, but not glans penis, contains a high density of fine-touch mechanoreceptors. Clinically the penilo-cavernosus reflex provides information on function of the sacral nerves. The study demonstrated that in the majority of circumcised men this reflex cannot be elicited clinically, but can be measured neurophysiologically.

OBJECTIVE

- To test clinical observations that the penilo-cavernosus reflex is much more difficult to elicit in circumcised men.

PATIENTS AND METHODS

- Men consecutively referred for uro-neurological or uro-neurophysiological examination were prospectively included.
- Those with possible sacral neuropathic lesions were excluded.
- A history was obtained, and a clinical neurological examination was performed.
- The penilo-cavernosus reflex was tested clinically and neurophysiologically using electrical and mechanical stimulation.
- Reflex elicitability scores in groups of circumcised men, men with foreskin retraction and a control group of uncircumcised men were compared using the Mann–Whitney *U* test.

RESULTS

- The reflex was clinically non-elicitable in 73%, 64% and 8% of 30 circumcised men, 15 men with foreskin retraction, and 29 control men, respectively.
- The scored reflex elicitability was significantly ($P < 0.001$) higher in control men than in the other two groups clinically, but not neurophysiologically.

CONCLUSION

- The study confirmed the lower clinical and similar neurophysiological elicitability of the penilo-cavernosus reflex in circumcised men and in men with foreskin retraction. This finding needs to be taken into account by urologists and other clinicians in daily clinical practice.

KEYWORDS

bulbocavernosus reflex, circumcision, foreskin, neuropathic lesion, penis

INTRODUCTION

In patients with a suspected neurogenic cause of bladder, bowel or sexual dysfunction, sacral (i.e. bulbocavernosus) reflex testing is often useful. Such testing assesses the continuity and function of the sacral reflex arc, which includes the afferent sensory neuron, segmental spinal reflex pathway, and the sacral lower motor neuron [1]. Sacral reflexes can be tested both clinically and neurophysiologically. Clinical testing is particularly useful in men [2–5], and it has been shown to provide important information that is not always revealed by other sacral clinical testing (e.g. assessment of perianal sensation) [6]. On performing sacral reflex testing in daily clinical practice it was, however, my impression that the penilo-cavernosus reflex [7] is much more difficult to elicit in circumcised men, even when they have no other clinical signs indicative of a possible sacral neurogenic lesion [6,8,9]. A similar observation was also made in non-circumcised men with permanent retraction of the foreskin behind the glans penis.

To test this observation, groups of circumcised men and men with foreskin retraction were recruited and the clinical elicitability of the penilo-cavernosus reflex in these two groups was compared with the clinical elicitability in a control group of men. Furthermore, in most subjects, the penilo-cavernosus reflex was also measured neurophysiologically.

PATIENTS AND METHODS

In the period from May 2003 to December 2009 all consecutive circumcised men and all men with foreskin retraction referred to me for uro-neurological or uro-neurophysiological examination were prospectively included. From the same population, but over a shorter period of time, uncircumcised control men were also recruited. All subjects were referred because of minor sacral dysfunctions (e.g. inguinal pain, voiding or defecation problems). Those with a history suggestive of possible focal sacral neuropathy (sciatica, lumbosacral spine or pelvic fracture or surgery), peripheral neuropathy (diabetes, renal failure, alcohol abuse) or extrapyramidal disorders, and patients reporting perineal

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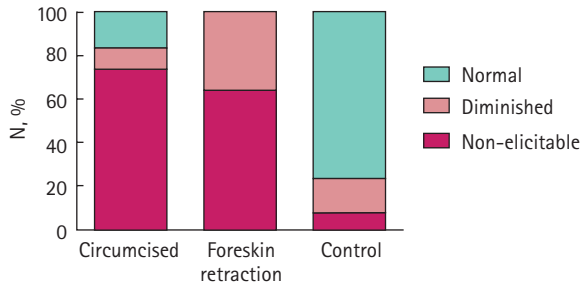


FIG. 1.
Percentage of circumcised men, men with foreskin retraction, and men in a control group with clinically non-elicitable, diminished or normal penilo-cavernosus reflexes.

numbness or paraesthesia, were excluded from all three groups. Furthermore, all subjects with abnormalities on clinical neurological examination of the sacral region were excluded. The study was approved by the National Ethics Committee of Slovenia, and all subjects provided informed consent.

In all patients a history of the current disease was obtained, with particular emphasis on sacral (urinary, bowel and sexual) dysfunction. Clinical neurological examination of the trunk and lower limbs was performed, focusing on muscle atrophy, tone and power (including anal sphincter squeeze), sensation (including saddle touch and pinprick), myotatic reflexes, the penilo-cavernosus reflex, and plantar responses. The penilo-cavernosus reflex was elicited with the men in the dorsal lithotomy position by brisk compression of the glans penis between the first three fingers. The foreskin was left in position over the glans in the control group of men, and behind the glans in men presenting with foreskin retraction (i.e. no foreskin manipulation was attempted). After applying a stimulus, movement of the perineum was carefully observed, and if an unequivocal response could be seen, the reflex was regarded as normal (scoring 2). The reflex was diminished when a reproducible and unequivocal movement could not be seen, but could be felt with the fingers of the contralateral hand applied against the patient's perineum (scoring 1). If no response could be either seen or felt, the reflex was regarded as clinically non-elicitable (scoring 0) [10].

In addition, neurophysiological measurement of the sacral reflex was performed using an electromyography (EMG) system (Keypoint; Alpine Biomed Neurodiagnostics, Skovlunde, Denmark) with standard settings (filters, 10 Hz to 10 kHz) [11]. Single-pulse and double-pulse electrical stimulation of the dorsal penile nerves was applied using a hand-held surface electrode assembly [12,13], and mechanical stimulation of the glans penis using an electromechanical hammer [10,12,14]. Responses were recorded by a standard concentric EMG needle electrode inserted consecutively into the left and right bulbocavernosus muscles [10,15]. Apart from noting the presence or absence of responses at the maximal tolerated stimulation intensity, latency of the responses was also measured and compared with our confidence intervals (<39 ms on single-pulse electrical stimulation, <36 ms on double-pulse electrical stimulation, and <35 ms on mechanical stimulation) [9]. A neurophysiologically non-elicitable penilo-cavernosus reflex scored 0, a reflex with latency prolongation scored 1, and a normal latency reflex scored 2 for further statistical analyses.

Penilo-cavernosus reflex elicitability scores on clinical and neurophysiological testing, and patients' ages in all three groups of men were compared using the Mann–Whitney *U* test, and the *Z*-test for two proportions [16].

RESULTS

In the study period I performed uro-neurophysiological examination in 247 men. Of these, 31 were circumcised and 15 had retraction of the foreskin. In addition, three circumcised men and one man with foreskin retraction who only underwent uro-neurological examination were included. However, four of the 34 circumcised men, and one of the 16 men with foreskin retraction were excluded because of findings in their history or on clinical examination indicating a possible sacral neurogenic lesion. Finally, 74 men were included in the study. The group of circumcised men consisted of 30 subjects, aged 8–82 years (mean ± SD, 48 ± 15 years), the group with foreskin retraction included 15 men, aged 26–74 years (56 ± 17 years), and the control group comprised 29 men, aged 32–83 years (52 ± 14 years). No significant difference in age was found between the three studied groups using the Mann–Whitney *U* test.

The clinical elicitability of the penilo-cavernosus reflex in all three studied groups is shown in Fig. 1. The reflex was normal in 5, 0 and 20, diminished in 3, 5 and 4, and non-elicitable in 22, 9 and 2 circumcised men, men with foreskin retraction, and control men, respectively. The difference in scored elicitability of the reflex was highly significant ($P < 0.001$) between pairs of circumcised and control men and of men with foreskin retraction and control men. In contrast, no significant difference in elicitability of the reflex was found on comparison of circumcised men and men with foreskin retraction on the Mann–Whitney *U* test ($P = 0.85$).

Neurophysiological measurement of the penilo-cavernosus reflex using electrical and mechanical stimulation was performed in 19 of 30 (63%) circumcised men, 12 of 15 (80%) men with foreskin retraction, and in 27 of 29 (93%) control men. The reflex could not be detected in one circumcised man bilaterally, and was of prolonged latency in one control subject unilaterally. In all other tested men included in this study the penilo-cavernosus reflex could be elicited and was of normal latency on neurophysiological testing. No significant difference between all three studied groups of men was found on comparison of scored neurophysiological measurements of the penilo-cavernosus reflex on the *Z*-test for two proportions ($P > 0.25$).

DISCUSSION

The present study confirmed my previous observations [6,8,9] that the penilo-cavernosus reflex is more difficult to elicit clinically in circumcised men. Furthermore, the study also showed a similar result in men with persistent foreskin retraction. This finding suggests that a clinically non-elicitable or diminished penilo-cavernosus reflex in a circumcised man or in a man with persistent foreskin retraction is of lower clinical significance (i.e. positive

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predictive value for sacral neurogenic lesion) than in a man without either of these two conditions. This finding needs to be taken into account by physicians using penilo-cavernosus reflex testing in daily clinical practice.

The probable reason for this finding in circumcised men is the elimination of the most sensitive part of the penis (i.e. the foreskin), and to a lesser extent, desensitization of sensory receptors in the penile glans [17,18]. Histological studies have shown mainly free nerve endings responding to deep pressure and pain in the glans penis [19], and a high density of fine-touch mechanoreceptors (e.g. Meissner's corpuscles) in the transitional area from the external to the internal surface (i.e. ridged band) and the frenulum of the foreskin [20]. It seems that the brisk finger grip used during clinical testing of the penilo-cavernosus reflex is an inadequate mechanical stimulus to activate the required number of receptors in the denuded and desensitized glans penis to elicit a reflex in a typical circumcised man. By contrast, as shown in this study, the mechanical stimulus provided by the electromechanical hammer is capable of eliciting an appropriate reflex in most circumcised men. It seems that the electromechanical hammer provides a stronger and more robust stimulus, which is able to activate the somewhat desensitized receptors of the glans penis itself. However, electrophysiological recording of the reflex using a needle EMG electrode inserted into the muscle may also be more sensitive in detecting a response [2,3,5,10].

The population of men with 'foreskin retraction' has not, to my knowledge, been specifically mentioned previously in the literature. All non-circumcised men presenting with a completely exposed glans penis were included in this group. With regard to penilo-cavernosus reflex elicitability, they appear to be no different to circumcised men.

Although the present study was not designed to obtain exact epidemiological information, 31 (13%) circumcised men, and 14 (6%) men with foreskin retraction out of 247 consecutively referred men with minor sacral dysfunction provide a crude impression of their proportions in the middle-aged male population in Slovenia. The rather low percentage of circumcised men contrasts with the situation in many other countries, notably USA, Israel and Muslim countries, where infant male circumcision is the most common medical procedure. With the high percentage of circumcised men in the USA population in mind, it is interesting to note that Blaivas *et al.* [5] were still able to clinically elicit the penilo-cavernosus reflex in virtually all (98%) 'normal' men. With clinical elicitation, and detection of the reflex by an EMG needle inserted into the external urethral sphincter muscle, they reported detection in all (100%) of control men [5]. They detected the reflex by placing the thumb of the examining hand against the anus and simultaneously palpating the bulbocavernosus and ischiocavernosus muscles with the index and middle fingers. Their technique of reflex detection might possibly increase the percentage of clinically detected reflex responses. Unfortunately, they did not mention the percentage of circumcised control men tested in their study [5]. Other authors from the USA have reported lower percentages of controls with a clinically elicitable penilo-cavernosus reflex (e.g. 70% [3], 86% [2]), but the percentage of circumcised patients in their populations is also not known.

The physiological function of the penilo-cavernosus reflex has not been studied in detail. In contrast to myotatic reflexes (e.g. knee or Achilles jerks), which are primarily responses to artificial and physiologically unimportant stimuli, the penilo-cavernosus reflex is assumed to have the physiological function of assisting in emptying the urethra after voiding, and of being part of the ejaculation process. It is interesting to link the results of the present study to studies assessing the impact of circumcision on sexual function in adult men [21,22].

Premature ejaculation was suggested, by some studies measuring vibratory threshold, to be the result of increased sensitivity of the penile glans [23]. This view was further supported by improvement in intravaginal ejaculation latency time and other outcome measures in patients with premature ejaculation after application of the topical desensitizing agent PSD502 (i.e. aerosolized lidocaine–prilocaine) [24]. Circumcision might have a similar desensitizing effect on the penile glans, and may be an alternative therapeutic approach for patients with premature ejaculation. This view is supported by a small study including 22 patients circumcised for phimosis (11 patients), balanitis (four patients), condyloma (three patients) or aesthetics (three patients). After the procedure, frequency of premature ejaculation was reduced in this group from seven to three patients [22].

The limitation of the study is that, as the only examiner in the study, I was non-blinded for subjects' clinical status, as well as for their foreskin status. The clinical status should not have affected the results as all men with suspected neurogenic lesions were excluded. Regarding non-blinded foreskin status, it is very difficult to imagine a study design to eliminate this problem. Nevertheless, I do not think these limitations were significant as the results were clear-cut. An additional limitation was the inclusion of men with minor sacral symptoms. However, subjects from all three groups were recruited in an identical way, from the same patient population. Furthermore, included patients did not have characteristic neurological symptoms and any abnormal findings on neurological examination. The inclusion of patients referred for minor sacral dysfunction is also clinically more relevant than inclusion of subjects with no symptoms, who are not expected to present to a medical service [25].

The present study showed the lower utility of the clinically elicited penilo-cavernosus reflex in circumcised men and in men with persistent foreskin retraction. This means that a clinically non-elicitable or diminished penilo-cavernosus reflex in this population should be interpreted with caution. To exclude a sacral neurogenic lesion in circumcised men and in men with foreskin retraction, neurophysiological testing may be indicated more often. The findings presented in this study might also explain the possible difference in sexual function between these two populations of men.

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CONFLICT OF INTEREST

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Correspondence: Simon Podnar, Institute of Clinical Neurophysiology, University Medical Centre Ljubljana, SI-1525 Ljubljana, Slovenia. e-mail: simon.podnar@kclj.si

Abbreviation: **EMG**, electromyography.

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